

Differential drive · 2 drive wheels Ø87 · 4 casters · footprint 250 × 250 × 100 mm



1. All dimensions in millimetres (mm). Do not scale from print.
2. Datum: X from longitudinal centreline ($\pm$), Y from FRONT edge.
3. Drive axle at Y=160; TRACK (wheel c/c) = 220; wheel $\varnothing$ 87.
4. Caster positions dimensioned on PLAN & listed in CASTER LOCATION TABLE (right).
5. Keep centre of mass within Y=160–180 $\rightarrow$ ~60–80% static load on drive wheels.
6. Robot rests on 2 drive + 2 rear casters; front casters 1–2 mm proud to avoid over-constrained 6-point rocking. Mount battery as low as possible (low CoG).
7. Max sweep radius on spot-turn $\approx$ 203 (front corners) $\rightarrow$ swept $\varnothing \approx$ 406.

Technical drawing of a rectangular container. The container has a height of 100 and a width of 43.5. It is labeled "FRONT" on the left and "REAR" on the right. A large black wheel with a diameter of $\varnothing 87$ is positioned inside the container. Two small white wheels are positioned outside the container, one at the front and one at the rear, resting on a hatched ground line.

#	QTY	DESCRIPTION
1	2	Drive wheel Ø87 (differential, motor-driven)
2	2	Caster — REAR (load-bearing, spring optional)
3	2	Caster — FRONT (anti-tip, mount 1-2 mm high)

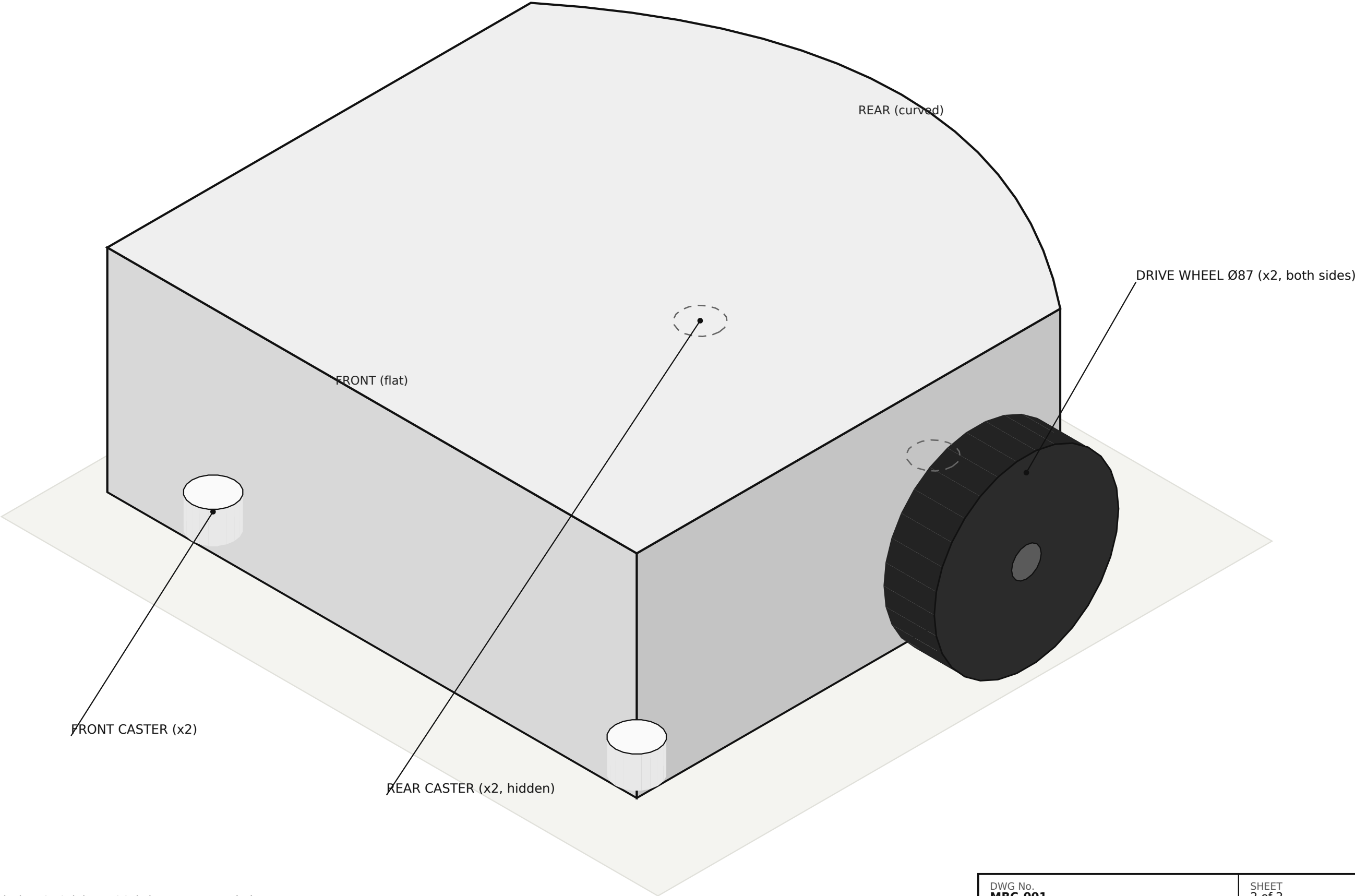
REF	X from CL (mm)	Y from FRONT (mm)	CASTER
C1	-100	25	Ø20 front *
C2	+100	25	Ø20 front *
C3	-55	210	Ø20 rear
C4	+55	210	Ø20 rear

Datum: X from centreline ($\pm$), Y from FRONT edge. * front casters mounted 1–2 mm proud. Working height = GC 15.

TITLE Wheel Layout — Diff-Drive Chassis		DRAWN BY UCHIDA	
SCALE 1:2	UNITS mm	DATE 2026-07-17	SIZE A3
DWG No. MRC-001		PROJECTION 3rd-angle	REV A

MOBILE ROBOT CHASSIS — ISOMETRIC VIEW

Illustrative 3D · front-right-top iso · dimensions per Sheet 1 (MRC-001)



Overall envelope: 250 (W) × 250 (L) × 100 (H) mm · ground clearance 15 mm
Left drive wheel & rear casters shown dashed (hidden). Full dimensions on Sheet 1.

DWG No. MRC-001	SHEET 2 of 2
VIEW Isometric	SCALE NTS